Solution

According to Joule's heating law $H = I^2Rt$

The current passed through the electrical heater $= \frac{220\text{V}}{10\Omega} = 22\,\text{A}$

Heat produced in one second by the electrical heater $H = I^2R$

Heat produced in one second
$H = (22)^2 \times 10 = 4840$ J = 4.84 k J. In fact the power rating of this electrical heater is 4.84 k W.

The amount of heat energy to increase the temperature of 1kg water from 30°C to 60°C is

$Q = ms\,\Delta T$ (Refer XI physics vol 2, unit 8)

Here $m = 1$ kg,
$s = 4200$ J kg^{-1} K^{-1},
$\Delta T = 30$ K,

so $Q = 1 \times 4200 \times 30 = 126$ kJ

The time required to produce this heat energy $t = \frac{Q}{I^2R} = \frac{126 \times 10^3}{4840} \approx 26.03$ s

2. Electric fuses

Fuses as shown in Figure 2.31, are connected in series in a circuit to protect the electric devices from the heat developed by the passage of excessive current. It is a short length of a wire made of a low melting point material. It melts and breaks the circuit if current exceeds a certain value. An alloy of lead - tin is used for fuses when current rating is below 15 A and when current rating is above 15 A, copper fuse wires are used.

The only disadvantage with the above fuses is that once fuse wire is burnt due to excessive current, they need to be replaced. Nowadays in houses, circuit breakers

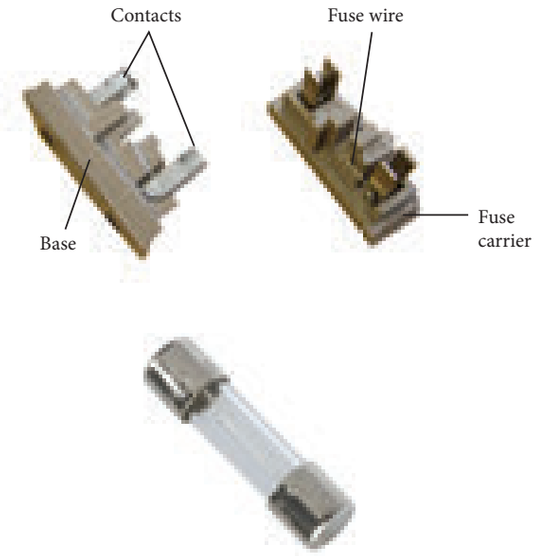


Figure 2.31 Electric Fuse

(trippers) are also used instead of fuses. Whenever there is an excessive current produced due to faulty wire connection, the circuit breaker switch opens. After repairing the faulty connection, we can close the circuit breaker switch. It is shown in the Figure 2.32.



Figure 2.32 circuit breakers

3. Electric furnace

Furnaces as shown in Figure 2.33 are used to manufacture a large number of technologically important materials such as steel, silicon carbide, quartz, gallium arsenide, etc. To produce temperatures up to 1500°C, molybdenum-nichrome wire wound on a silica tube is used. Carbon arc furnaces produce temperatures up to 3000°C.

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